

5. Bond Price Volatility and Interest Rate Risk

1. Calculate the requested measures in parts (a) through (f) for bonds A and B (assume that each bond pays interest semiannually):

	Bond A	Bond B
Coupon	8%	9%
Yield to maturity	8%	8%
Maturity (years)	2	5
Par	\$100.00	\$100.00
Price	\$100.00	\$104.055

(a) What is the price value of a basis point for bonds A and B?

For **bond A**, we get a bond quote of \$100.00 for our initial price if we have a 2-year maturity, an 8% coupon rate and an 8% yield. If we change the yield one basis point so the yield is 8.01%, then we have the following variables and values:

$$C = \$40, y = 0.0801 / 2 = 0.04005 \text{ and } n = 2(2) = 4.$$

Inserting these values into the present value of the coupon payments formula, we get:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$40 \left[\frac{1 - \frac{1}{(1.04005)^4}}{0.04005} \right] = \$145.179.$$

Computing the present value of the par or maturity value of \$1,000 gives:

$$\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.04005)^4} = \$854.640.$$

If we add a basis point to the yield, we get the value of Bond A as: $P = \$145.179 + \$854.640 = \$999.819$ with a bond quote of \$99.9819. For **bond A** the price value of a basis point is about $\$100 - \$99.9819 = \$0.0181$ per \$100.

Using the bond valuation formulas as just completed above, the value of **bond B** with a yield of 8%, a coupon rate of 9%, and a maturity of 5 years is: $P = \$364.990 + \$675.564 = \$1,040.554$ with a bond quote of \$104.0554. If we add a basis point to the yield, we get the value of Bond B as: $P = \$364.899 + \$675.239 = \$1,040.139$ with a bond quote of \$104.0139. For **bond B**, the price value of a basis point is $\$104.0554 -$

\$104.0139 = **\$0.0416** per \$100.

(b) Compute the Macaulay durations for the two bonds.

For **bond A** with $C = \$40$, $n = 4$, $y = 0.04$, $P = \$1,000$ and $M = \$1,000$, we have:

$$\text{Macaulay duration (half years)} = \frac{\frac{1C}{(1+y)^1} + \frac{2C}{(1+y)^2} + \dots + \frac{nC}{(1+y)^n} + \frac{nM}{(1+y)^n}}{P} =$$

$$\frac{\frac{1(\$40)}{(1.04)^1} + \frac{2(\$40)}{(1.04)^2} + \dots + \frac{4(\$40)}{(1.04)^4} + \frac{4(\$1,000)}{(1.04)^4}}{\$1,000} = \frac{\$3,775.09}{\$1,000} = 3.77509.$$

Macaulay duration (years) = Macaulay duration (half years) / 2 = 3.77509 / 2 = **1.8875**.

For **bond B** with $C = \$45$, $n = 10$, $y = 0.04$, $P = \$1,040.55$ and $M = \$1,000$, we have:

$$\text{Macaulay duration (half years)} = \frac{\frac{1C}{(1+y)^1} + \frac{2C}{(1+y)^2} + \dots + \frac{nC}{(1+y)^n} + \frac{nM}{(1+y)^n}}{P} =$$

$$\frac{\frac{1(\$45)}{(1.04)^1} + \frac{2(\$45)}{(1.04)^2} + \dots + \frac{10(\$45)}{(1.04)^4} + \frac{10(\$1,000)}{(1.04)^4}}{\$1,040.55} = \frac{\$8,645.2929}{\$1,040.55} = 8.3084.$$

Macaulay duration (years) = Macaulay duration (half years) / 2 = 8.3084 / 2 = **4.1542**.

(c) Compute the modified duration for the two bonds.

Taking our answer for the Macaulay duration in years in part (b), we can compute the modified duration for **bond A** by dividing by 1.04. We have:

$$\text{modified duration} = \text{Macaulay duration} / (1+y) = 1.8875 / 1.04 = \mathbf{1.814948}.$$

Taking our answer for the Macaulay duration in years in part (b), we can compute the modified duration for **bond B** by dividing by 1.04. We have:

$$\text{modified duration} = \text{Macaulay duration} / (1+y) = 4.1542 / 1.04 = \mathbf{3.994417}.$$

(d) Compute the approximate duration for bonds A and B using the shortcut formula by changing yields by 20 basis points and compare your answers with those calculated in part (c).

To compute the approximate measure for **bond A**, which is a 2-year 8% coupon bond trading at 8% with an initial price (P_0) of \$1,000 (thus, it trades at its par value of \$1,000), we proceed as follows.

First, we increase the yield on the bond by 20 basis points from 8% to 8.20%. Thus, Δy is 0.20% or 0.002 in decimal form. The new price (P_+) can be computed using our bond valuation formula. Doing this we get \$996.379 with a bond price quote of \$99.6379.

Second, we decrease the yield on the bond by 20 basis points from 8% to 7.8%. The new price (P_-) can be computed using our bond valuation formula. Doing this we get \$1,003.638 with a bond price quote of \$100.3638.

Third, with the initial price, P_0 , equal to \$100 (when expressed as a bond quote), the duration can be approximated as follows:

$$\text{approximate duration} = \frac{P_- - P_+}{2(P_0)(\Delta y)}$$

where Δy is the change in yield used to calculate the new prices (in decimal form). What the formula is measuring is the average percentage price change (relative to the initial price) per 20-basis-point change in yield. Inserting in our values, we have:

$$\text{approximate duration} = \frac{\$100.3638 - \$99.6379}{2(\$100)(0.002)} = \mathbf{1.814948}.$$

This compares with 1.814948 computed in part (c). Thus, the approximate duration measure (to six decimal places for this problem) is the same as the modified duration computed in part (c).

To compute the approximate measure for **bond B**, which is a 5-year 9% coupon bond trading at 8% with an initial price (P_0) of \$104.0554, we proceed as follows.

First, we increase the yield on the bond by 20 basis points from 8% to 8.20%. Thus, Δy is 0.002. The new price (P_+) can be computed using our bond valuation formula. Doing this we get \$1,032.283 with a bond price quote of \$103.2283.

Second, we decrease the yield on the bond by 20 basis points from 8% to 7.8%. The new price (P_-) can be computed using our bond valuation formula. Doing this we get

\$1,048.909 with a bond price quote of \$104.8909.

Third, with the initial price, P_0 , equal to \$104.0554 (when expressed as a bond quote), the duration can be approximated as follows:

$$\text{approximate duration} = \frac{P_- - P_+}{2(P_0)(\Delta y)}$$

where Δy is the change in yield used to calculate the new prices (in decimal form). What the formula is measuring is the average percentage price change (relative to the initial price) per 20-basis-point change in yield. Inserting in our values, we have:

$$\text{approximate duration} = \frac{\$104.8909 - \$103.2283}{2(\$104.0554)(0.02)} = \mathbf{3.994507}.$$

This compares with 3.994417 computed in part (c). Thus, the approximate duration measure is virtually the same as the modified duration computed in part (c).

(e) Compute the approximate convexity measure for bonds A and B using the shortcut formula by changing yields by 20 basis points.

To compute the approximate convexity measure for **bond A**, which is a 2-year 8% coupon bond trading at 8% with an initial price (P_0) of \$100, we proceed as follows.

First, we increase the yield on the bond by 20 basis points from 8% to 8.2%. Thus, Δy is 0.002. The new price (P_+) can be computed using our bond valuation formula. Doing this we get \$99.6379.

Second, we decrease the yield on the bond by 20 basis points from 8% to 7.8%. The new price (P_-) can be computed using our bond valuation formula. Doing this we get \$1,003.638 with a bond price quote of \$100.3638.

Third, with the initial price, P_0 , equal to \$100, the convexity measure of any bond can be approximated using the following formula:

$$\text{approximate convexity measure} = \frac{P_+ + P_- - 2P_0}{P_0(\Delta y)^2}.$$

Inserting in our values, the approximate convexity measure is

$$\text{approximate convexity measure} = \frac{\$100.3638 + \$99.6379 - 2(\$100)}{\$100(0.002)^2} = 4.2773486.$$

To compute the approximate convexity measure for **bond B** which is a 5-year 9% coupon bond trading at 8% with an initial price (P_0) of \$104.0554 (worth \$1,040.554 and with a par value = $M = \$1,000$), we proceed as follows.

First, we increase the yield on the bond by 20 basis points from 8% to 8.2%. Thus, Δy is 0.001. The new price (P_+) can be computed using our bond valuation formula. Doing this we get \$103.2283.

Second, we decrease the yield on the bond by 20 basis points from 8% to 7.8%. The new price (P_-) can be computed using our bond valuation formula. Doing this we get \$1,048.909 with a bond price quote of \$104.8909.

Third, with the initial price, P_0 , equal to \$104.0554, the convexity measure of any bond can be approximated using the following formula:

$$\text{approximate convexity measure} = \frac{P_+ + P_- - 2P_0}{P_0 (\Delta y)^2}.$$

Inserting in our values, the approximate convexity measure is:

$$\text{approximate convexity measure} = \frac{\$104.8909 + \$103.2283 - 2(\$104.0554)}{\$104.0554(0.002)^2} = 19.763824.$$

2. Can you tell from the following information which of the following three bonds will have the greatest price volatility, assuming that each is trading to offer the same yield to maturity?

Bond	Coupon Rate (%)	Maturity (years)
X	8	9
Y	10	11
Z	11	12

The price of a bond will change over time as a result of a change in the perceived credit risk of the issuer. Thus, if one of the three bonds undergoes greater change in credit risk then that bond might be expected to experience more volatility unless other factors

dominate. Below we describe these factors: term to maturity, coupon rate, and yield to maturity.

For a given term to maturity and initial yield, the price volatility of a bond will increase as the coupon rate becomes smaller. Thus, *ceteris paribus*, we would expect bond X to have greater price volatility than bond Y, and bond Y to have greater price volatility than bond Z. However, the differences in coupon rates for bonds Y and Z are not that great. Thus, if just looking at these two bonds, the differences in price volatilities may not be that recognizable.

For a given coupon rate and initial yield, longer terms to maturity will produce greater price volatility. Thus, *ceteris paribus*, bond Z will have more price volatility than bond Y and bond Y will have more price volatility than bond X. However, the differences in maturities for bonds Y and Z are not that great. Thus, if just comparing these two bonds, the differences in price volatilities may not be that identifiable.

We should note that the expectations of price volatility based upon coupon rates and maturities are the reverse for bonds X and Z. For example, we expect bond X to have the greatest price volatility based upon coupon rate but the lowest based upon maturity. For bond Z, we expect it to have lowest price volatility based upon coupon rate but the greatest based upon maturity. If the coupon rate and the maturity factors or characteristics balance out then it is possible all three bonds will experience price volatilities that are very similar.

3. State why you would agree or disagree with the following statement: If two bonds have the same dollar duration, yield, and price, their dollar price sensitivity will be the same for a given change in interest rates.

If the two bonds have the same dollar duration then their percentage change in price is the same. This implies they will have the same dollar price sensitivity. This possibility is seen from the following equation:

$$\frac{dP}{dy} = -(\text{modified duration})P$$

where the expression on the right-hand side is the estimated dollar duration. By having the same dollar duration, price (P), and yield, we see they can have the same price change (dP) for a given change in yield (dy). Thus, their dollar price change or dollar price sensitivity can be the same.

There are possible caveats to the above argument that make it possible that the dollar price sensitivity can be different for a given change in interest rates. For example, for an increase in the required yield, the estimated dollar price change is more than the actual price change. The reverse is true for a decrease in the required yield. Thus, to the extent the estimated changes can differ, their percentage changes in price can differ.

This makes it possible that their dollar price sensitivity will be different for a given change in interest rates. Also, coupon rates are a factor in determining price behavior. For an option-free bond with a given term to maturity and initial yield, the price volatility of a bond will increase as the coupon rate decreases. Thus, unless coupon rates are actually the same when the dollar duration, yield, and price are the same, then their dollar price sensitivity will not necessarily be the same for a given change in interest rates.

4. State why you would agree or disagree with the following statement: For a 1-basis point change in yield, the price value of a basis point is equal to the dollar duration.

The validity of the above statement is discussed below.

The price value of a basis point, also referred to as the dollar value of a 01, is the change in the price of the bond if the required yield changes by 1 basis point. For small changes in the required yield, the below equation does a good job in estimating the change in price:

$$dP = -(\text{dollar duration})(dy).$$

Consider a 6% 25-year bond selling at \$70.3570 to yield 9%. The dollar duration is 747.2009. For a 1-basis-point (0.0001) increase in the required yield, the estimated price change per \$100 of face value is

$$dP = -(\text{dollar duration})(dy) = -(\$747.2009)(0.0001) = -\$0.07472.$$

If we change the yield one basis point so the yield is 9.01%, then the value of the bond is: $P = \$592.378 + \$110.445 = \$702.824$ with a bond quote of \$70.2824. The price value of a basis point is about $\$70.2824 - \$70.3570 = -\$0.07464$. The dollar duration for a 1-basis point change gives about the same value as the price value of a basis point as both round off to **-\$0.0747**.

5. Consider the following two Treasury securities:

Bond	Price	Modified duration (years)
A	\$100	6
B	\$80	7

Which bond will have the greater dollar price volatility for a 25-basis-point change in interest rates?

The estimated dollar price change can be obtained by using the below equation:

$$dP = -(\text{modified duration})P(dy).$$

Inserting in our values for **bond A**, we have:

$$dP = -(\text{modified duration})P(dy) = -(6)\$100(0.0025) = -\$1.50$$

which is the estimated dollar price change or volatility for a 25-basis-point change. The percentage change in price is:

$$\frac{dP}{P} = \frac{\$1.50}{\$100} = -0.0150 \text{ or } -1.50\%.$$

Inserting in our values for **bond B**, we have:

$$dP = -(\text{modified duration})P(dy) = -(7)\$80(0.0025) = -\$1.40$$

which is the estimated dollar price change or volatility for a 25-basis-point change. The percentage change in price is:

$$\frac{dP}{P} = \frac{\$1.40}{\$80} = -0.0175 \text{ or } -1.750\%.$$

Thus, we see that while bond A has a greater estimated dollar price volatility compared to bond B, it has a lower percentage change in price. From an investor's point of view, every dollar invested in bond B has greater volatility.

6. Consider the following portfolio:

Bond	Market Value	Duration (years)
W	\$13 million	2
X	\$27 million	7
Y	\$60 million	8
Z	\$40 million	14

(a) What is the portfolio's duration?

The portfolio duration is equivalent to the weighted average of the duration for bond W (D_w), bond X (D_x), bond (D_y), and bond Z (D_z). We proceed as follows to calculate the portfolio duration.

First, we calculate the total market value (where M equal million). We have: total market value = $\sum$ market value of all four bonds = $MV_w + MV_x + MV_y + MV_z = \$13M$

$$+ \$27\text{M} + \$60\text{M} + \$40\text{M} = \$140\text{M}.$$

Second, we compute the portfolio weights as given by the following formula: weight (W) = market value of bond (MV) ÷ total market value (TMV) or $W_i = MV_i / TMV$ for $i = W, X, Y$ and Z . For the four weights we have:

$$\text{weight bond W} = W_w = MV_w / TMV = \$13\text{M} / \$140\text{M} = 13/140;$$

$$\text{weight bond X} = W_x = MV_x / TMV = \$27\text{M} / \$140\text{M} = 27/140;$$

$$\text{weight bond Y} = W_y = MV_y / TMV = \$60\text{M} / \$140\text{M} = 3/7;$$

$$\text{weight bond Z} = W_z = MV_z / TMV = \$40\text{M} / \$140\text{M} = 2/7.$$

The portfolio duration equals the weighted average of the duration for bond W (D_w), bond X (D_x), bond Y (D_y), and bond Z (D_z). We have:

$$(W_w)D_w + (W_x)D_x + (W_y)D_y + (W_z)D_z = (13/140)2 + (27/140)7 + (3/7)8 + (2/7)14 = \\ 0.1857142 + 1.350000 + 3.4285714 + 4.0000 = 8.9643 \text{ or } \mathbf{about\ 9}.$$

The portfolio's duration is 8.9643 and interpreted as follows: If all the yields affecting the four bonds in the portfolio change by 100 basis points, the portfolio's value will change by approximately 9%.

(b) If interest rates for all maturities change by 50 basis points, what is the approximate percentage change in the value of the portfolio?

The total change in value of a portfolio if all rates (for each point on the yield curve) change by the same number of basis points is simply the duration of a single security. Thus, we can proceed as follows when computing the approximate percentage change in the value of the portfolio by using the below formula where dy is 0.005 because all interest rates for all maturities change by 50 basis point. We have:

$$\frac{dP}{P} = -(\text{modified duration})(dy) = -(8.9643)(0.005) = -0.0448215 \text{ or about } \mathbf{-4.4821\%}.$$